

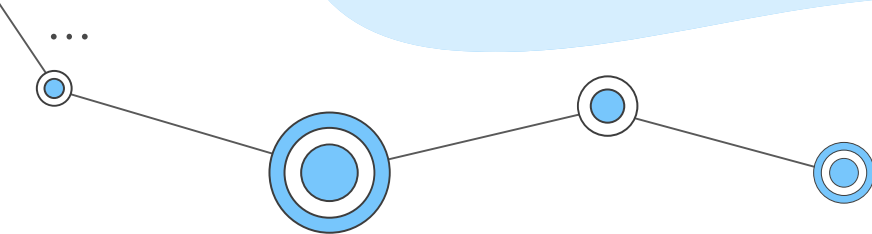
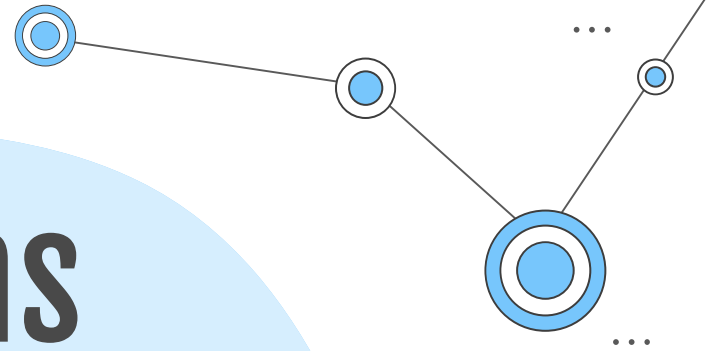
Regions

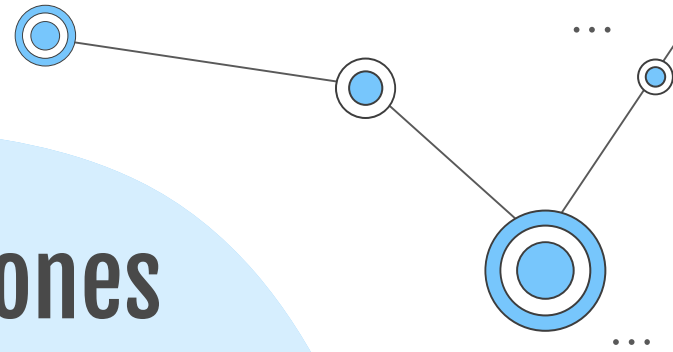
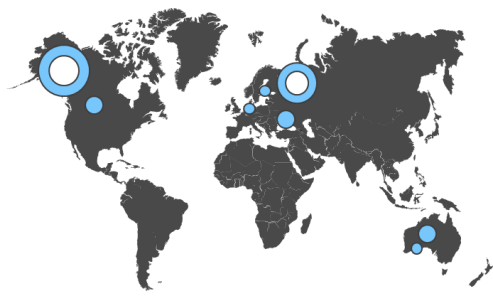
Region consists of one or multiple data centers
within a specific radius

Connected via low-latency network

Resources are deployed to regions

z.B. North Europe, West Europe, Germany West Central

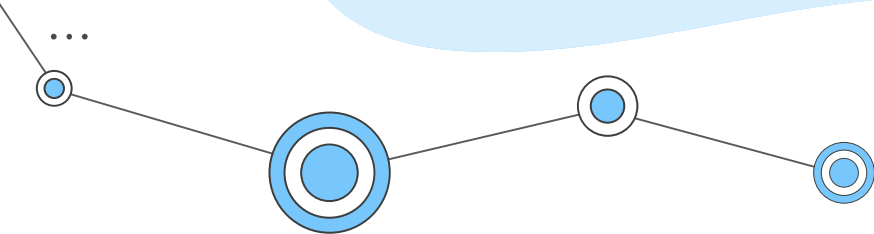


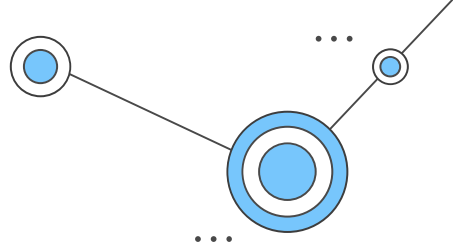


Availability Zones

Most of the regions support availability zones but not all of them

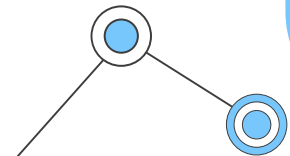
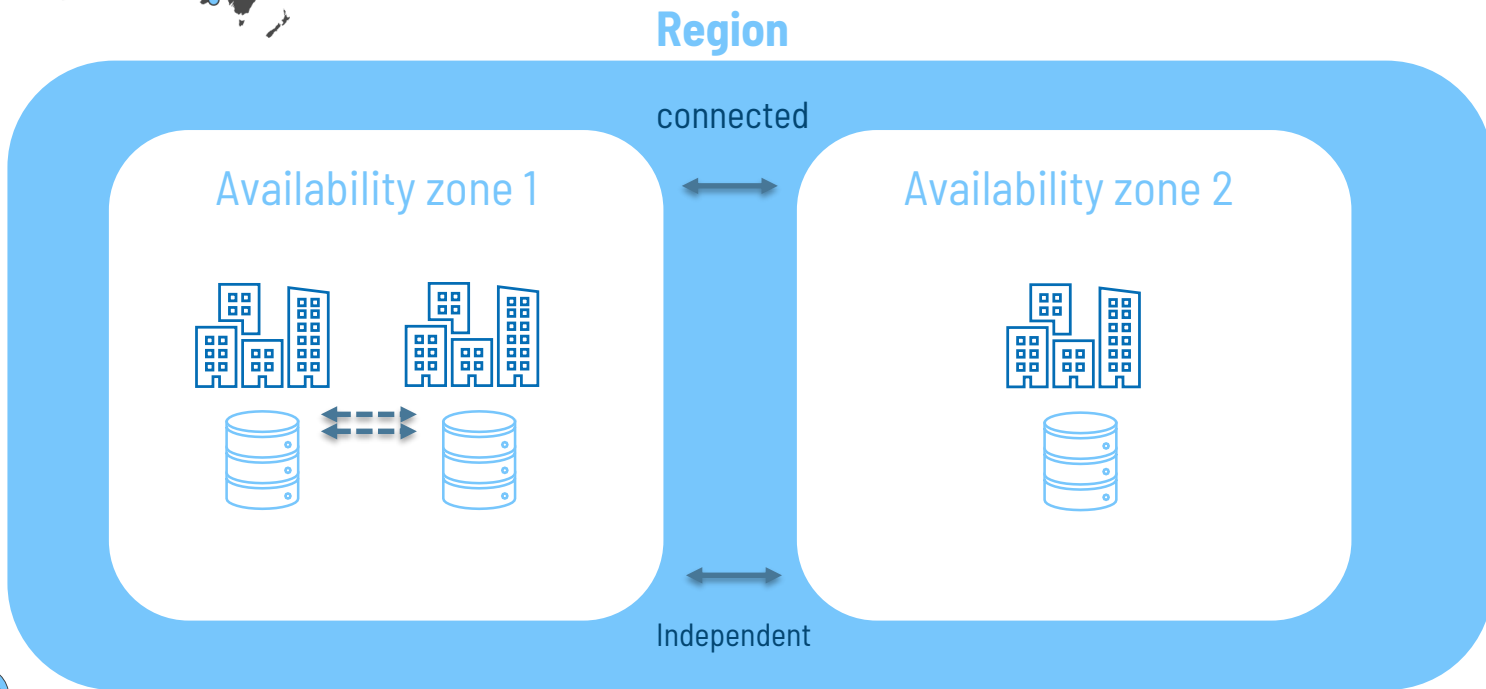
Designed to achieve redundancy and fault tolerance

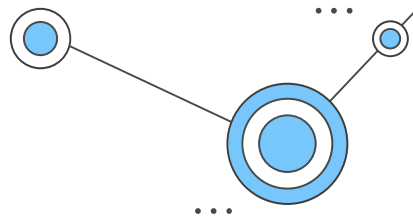




Availability zones

Physically separate locations within each Azure region

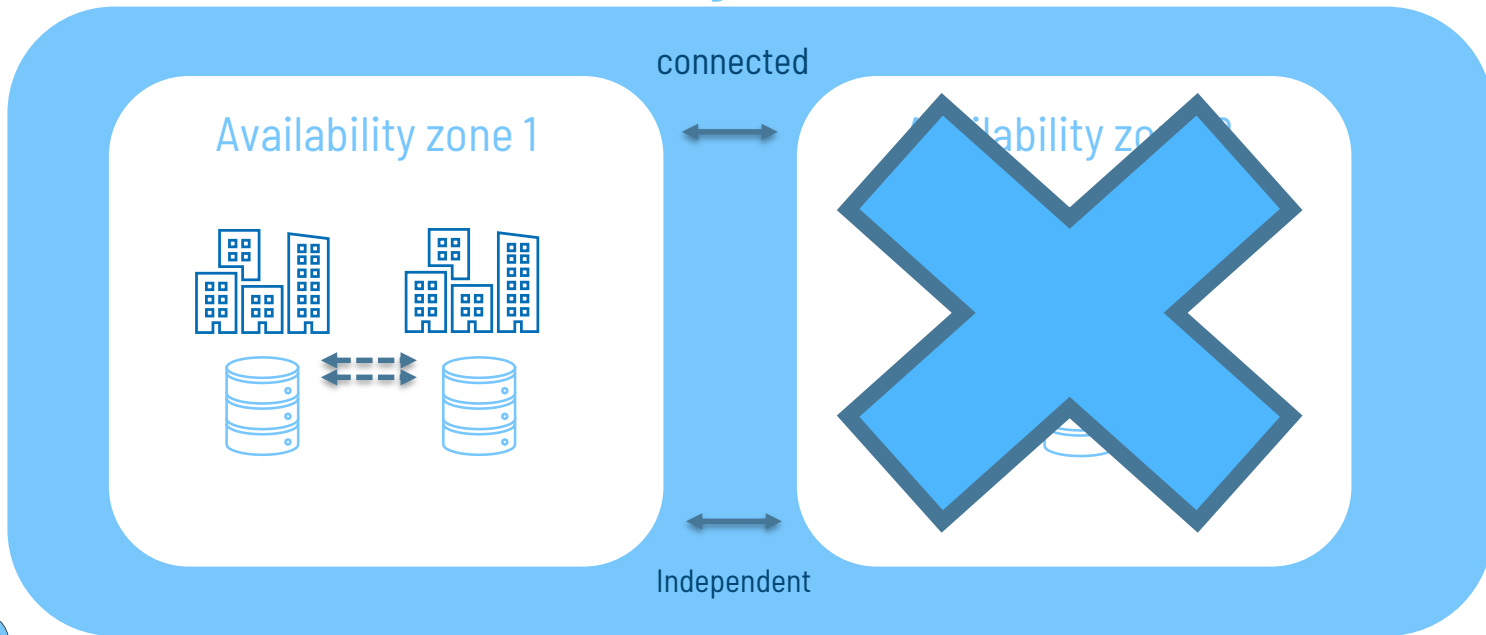




Availability zones

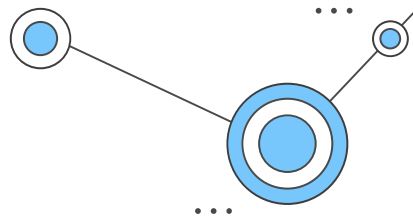
Physically separate locations within each Azure region

Region



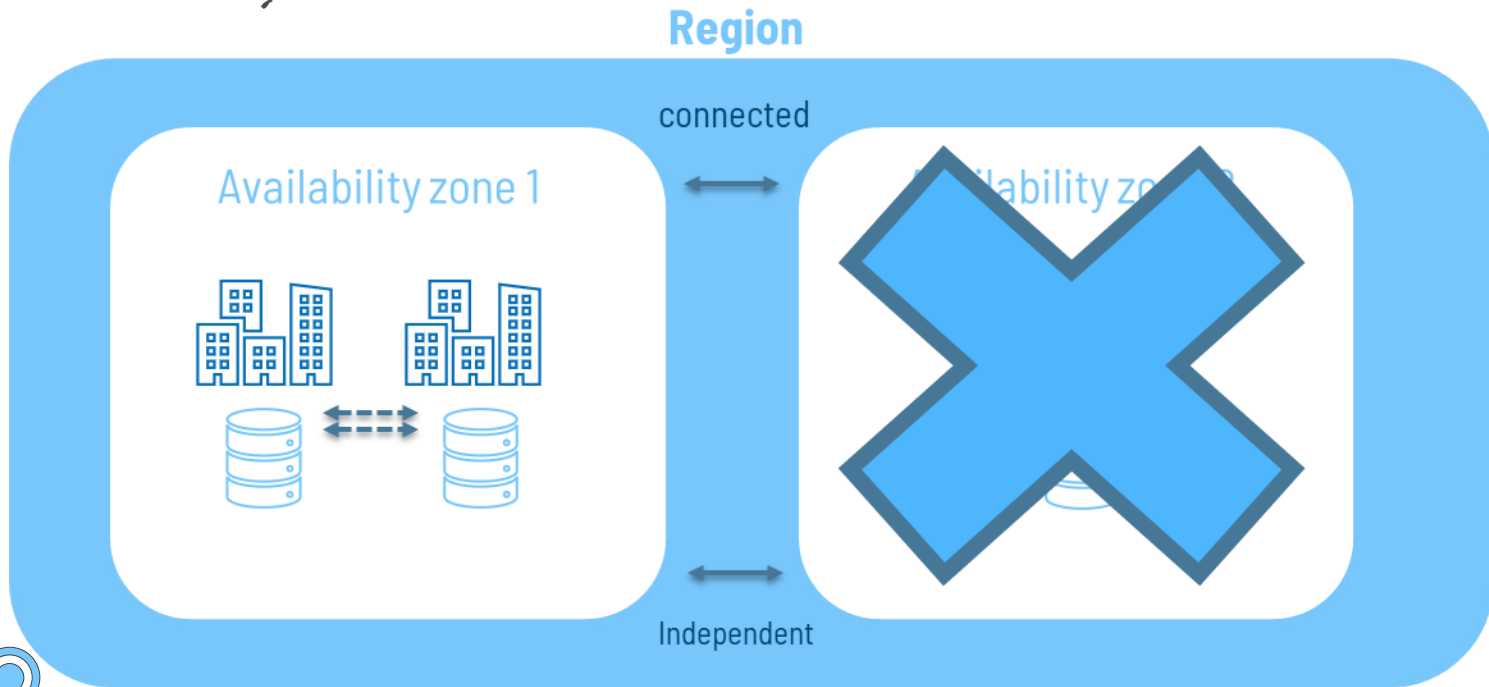
Availability zones

- ✓ **At least 3 availability zones** per availability zone supported region
- ✓ Connected via a **high-performance / low-latency network**
- ✓ **Independent power**, cooling, and networking infrastructure
- ✓ **Design to keep regions up** in the event of a disaster
- ✓ **Creating redundancies**



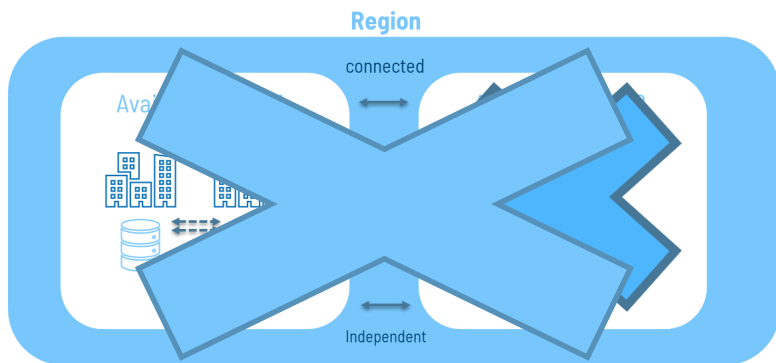
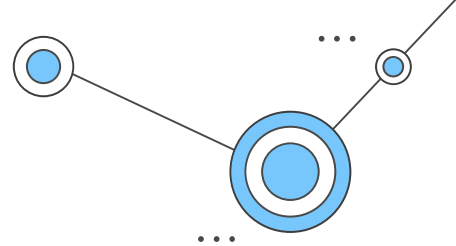
Availability zones

Physically separate locations within each Azure region



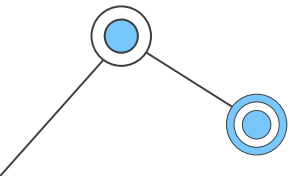


Availability zones



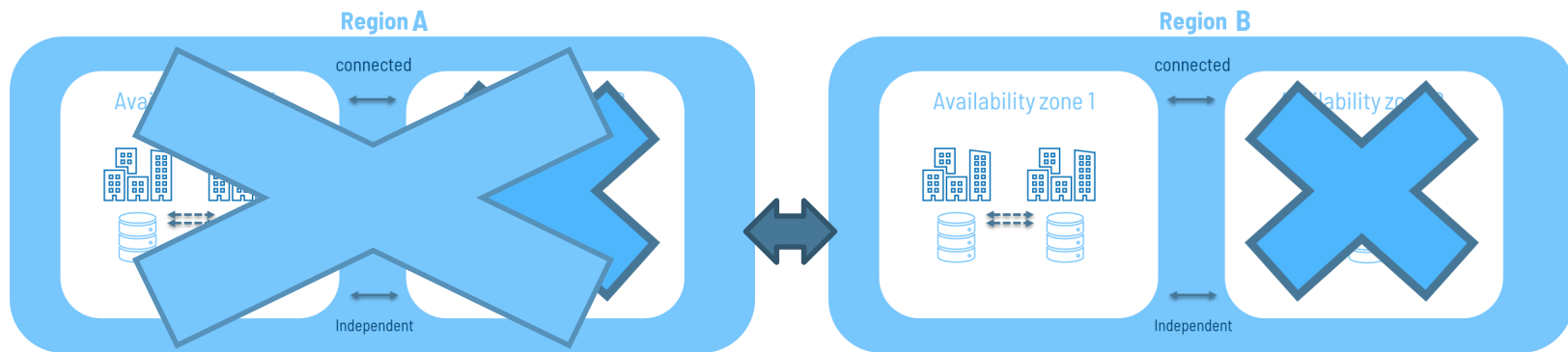
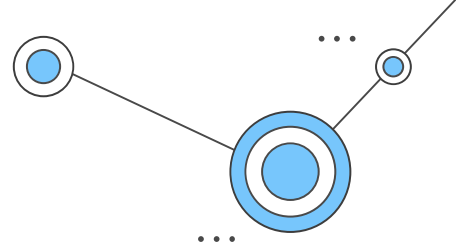
How to manage risks toward entire regions?

- Earthquakes,
- hurricanes,
- other natural or man-made disasters

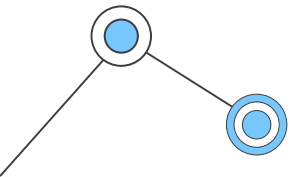


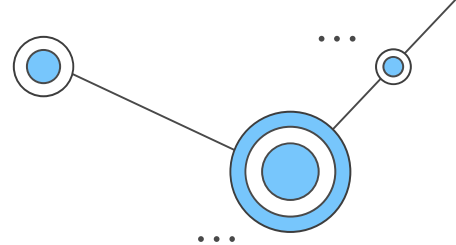


Availability zones

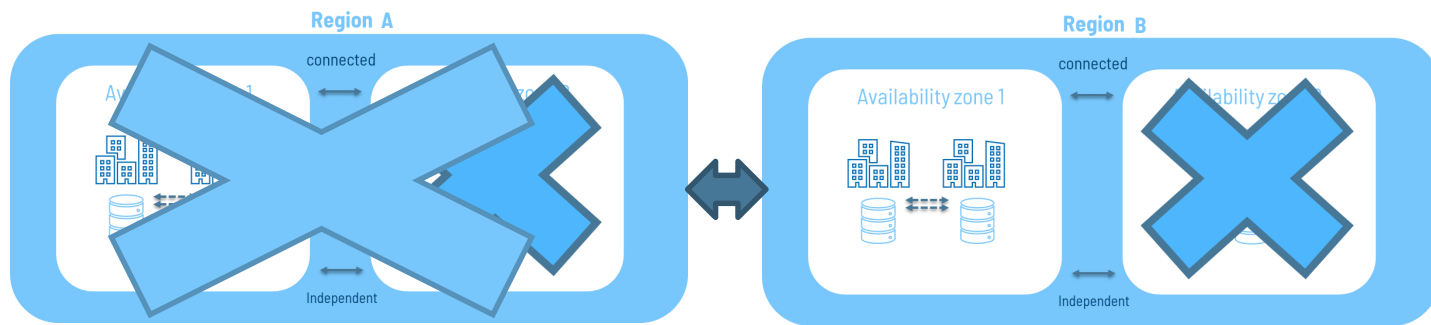


Geo-Replication





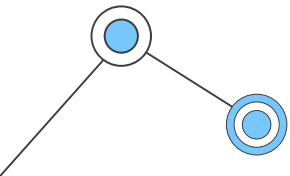
Availability zones



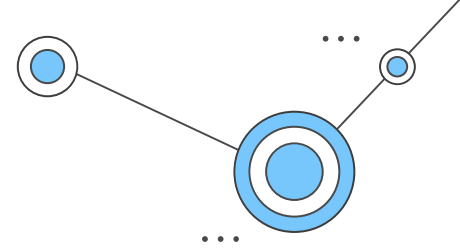
Geo-Replication

Sequential updates (only in one region at a time)

High availability & Reliability



Region pairs



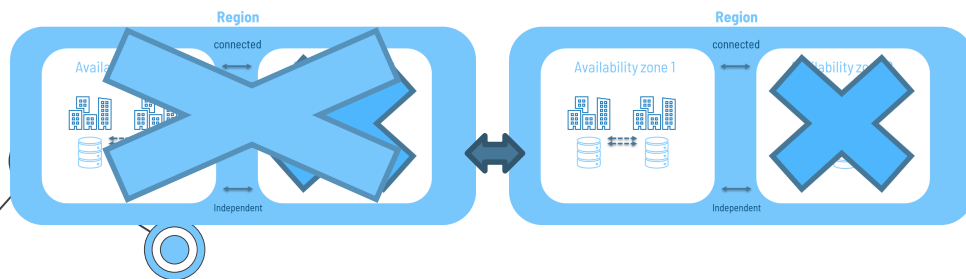
Factors involved in placement:

- Data residency
- Compliance
- Geo-political
- Internet latency

⇒ Region pairs: Recommend region for replication

Region pairs:

- Far enough to be isolated: > 300 miles
- Placed in the same geography



Geography

- Discrete market
- Maintaining data residency and compliance
- Withstand region failure

Region pairs



Geography	Region Pair A	Region Pair B
Asia Pacific	East Asia (Hong Kong)	Asia Southeast (Singapore)
Australia	Australia (East)	Australia Southeast
Australia	Australia, Central	Australia, Central 2*
Brazil	Brazil South	USA South Central
Brazil	Brazil, Southeast*	Brazil South
Canada	Canada, Central	Canada East
China	China North	China East
China	China North 2	China East 2
Europe	Europe North (Ireland)	West Europe (Netherlands)
France	France, Central	France South*
Germany	West-Central Germany	North Germany*
India	India, Central	India (South)
India	India, West	India (South)
Japan	Japan East	Japan, West
Korea	Korea, Central	Korea South
North America	US East	US West
North America	US East 2	US Central
North America	US North Central	US South Central
North America	US West 2	US West Central
Norway	Norway, East	Norway, West*
South Africa	South Africa, North	South Africa, West*
Switzerland	Switzerland, North	Switzerland, West*
UK	UK, West	UK, South
UAE	UAE, North	UAE, Central*
US Department of Defense	US DoD, East*	US DoD, Central*
US Government	US Gov Arizona*	US Gov Texas*
US Government	US Gov Iowa*	US Gov Virginia*
US Government	US Gov Virginia*	US Gov Texas*

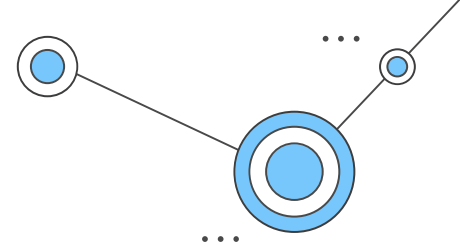
(*) Certain regions are subject to access restrictions

Sovereign Regions

Geography	Region Pair A	Region Pair B	...
Asia Pacific	East Asia (Hong Kong)	Asia Southeast (Singapore)	
Australia	Australia (East)	Australia Southeast	
Australia	Australia, Central	Australia, Central 2*	
Brazil	Brazil South	USA South Central	
Brazil	Brazil, Southeast*	Brazil South	
Canada	Canada, Central	Canada East	
China	China North	China East	
China	China North 2	China East 2	
Europe	Europe North (Ireland)	West Europe (Netherlands)	
France	France, Central	France South*	
Germany	West-Central Germany	North Germany*	
India	India, Central	India(South)	
India	India, West	India(South)	
Japan	Japan East	Japan, West	
Korea	Korea, Central	Korea South	
North America	US East	US West	
North America	US East 2	US Central	
North America	US North Central	US South Central	
North America	US West 2	US West Central	
Norway	Norway, East	Norway, West*	
South Africa	South Africa, North	South Africa, West*	
Switzerland	Switzerland, North	Switzerland, West*	
UK	UK, West	UK, South	
UAE	UAE, North	UAE, Central*	
US Department of Defense	US DoD, East*	US DoD, Central*	
US Government	US Gov Arizona*	US Gov Texas*	
US Government	US Gov Iowa*	US Gov Virginia*	
US Government	US Gov Virginia*	US Gov Texas*	

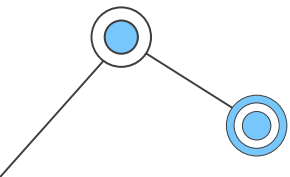
(*) Certain regions are subject to access restrictions

Sovereign Regions

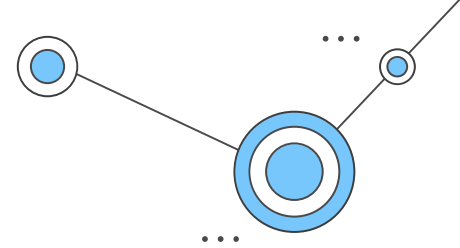


Separate from Public Cloud:

- Isolated from the main instance of Azure
- US Government, US DoD Central
- China
- Very strict compliance or legal requirements

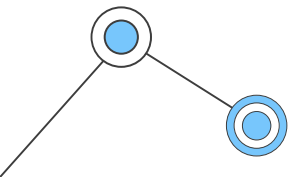


Sovereign Regions

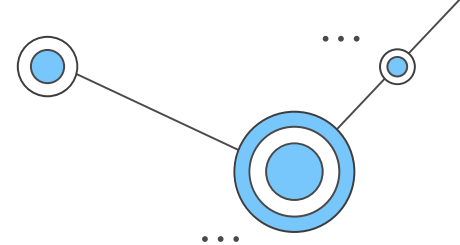


US Government

- Isolated from the main instance of Azure
⇒ <https://portal.azure.us>
- Meets the most complex compliance standards.
- Only US Government and its partners can use it
- From state to local government + partners
- Operated by screened U.S. personnel

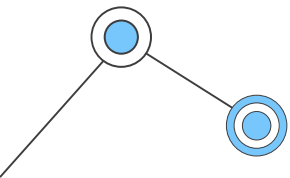


Sovereign Regions

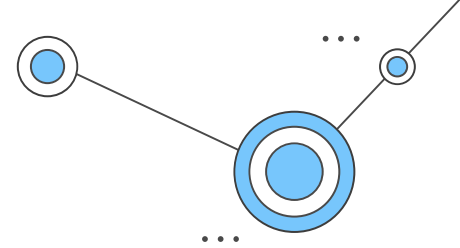


China

- More complex regulations to operate in China
- Data centers are not maintained directly by Microsoft
- Maintained by Microsoft partner: 21Vianet
- Separate instance: <https://portal.azure.cn>



Region pairs



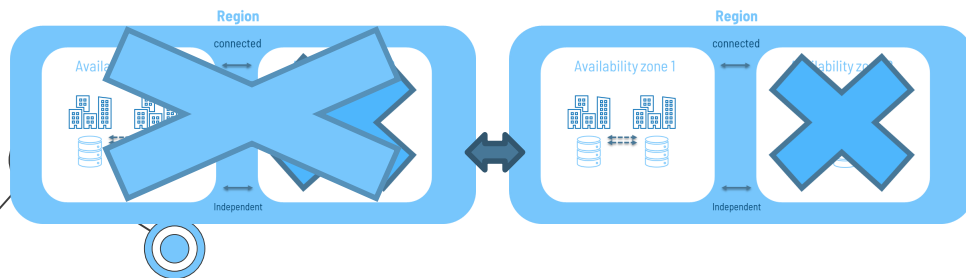
Factors involved in placement:

- Data residency
- Compliance
- Geo-political
- Internet latency

⇒ Region pairs: Recommend region for replication

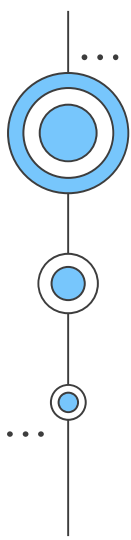
Region pairs:

- Far enough to be isolated: > 300 miles
- Placed in the same geography



Geography

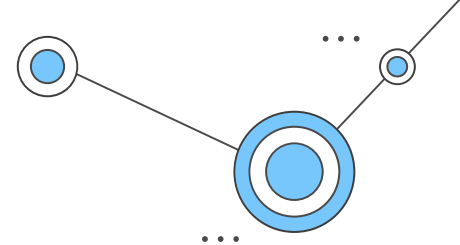
- Discrete market
- Maintaining data residency and compliance
- Withstand region failure



Resources & Resource groups



Resources & Resource groups



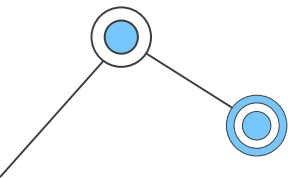
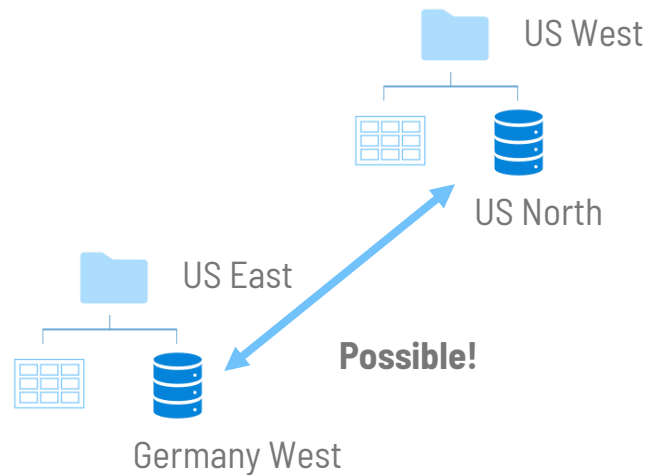
Resources

- Instances of services that you create
- Example: Virtual machines or SQL databases
- Deployed to a region



Resource group

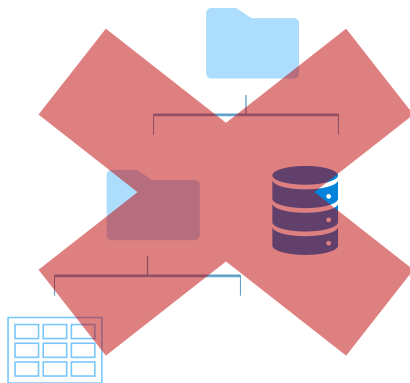
- Container of resources
- Management layer: Configure/delete resources
- Permissions are inherited (Tags are not inherited)
- Region of resource can be different from the region of resource groups
- Contains only metadata



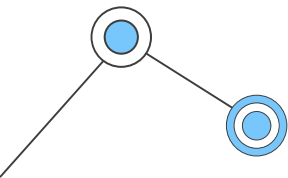
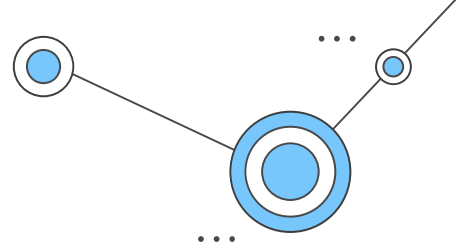
Resources & Resource groups

Resource group

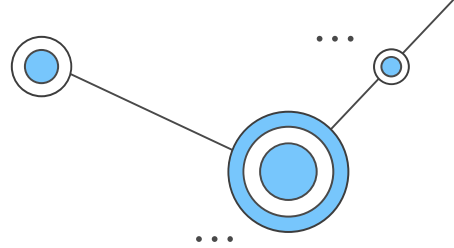
- Container of resources
- Management layer: Configure/delete resources
- Region of resource can be different from the region of resource groups
- Contains only metadata
- Cannot be nested



Not possible!

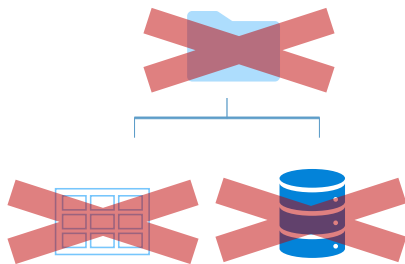


Resources & Resource groups

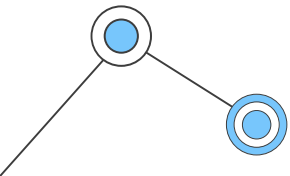


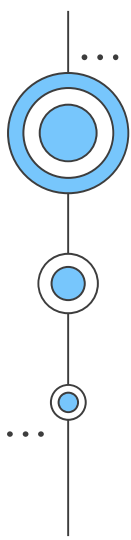
Resource group

- Container of resources
- Management layer: Configure/delete resources
- Region of resource can be different from the region of resource groups
- Contains only metadata
- Cannot be nested
- Deleting group deletes all contained resources



**Deletes ALL resources
in the resource group**

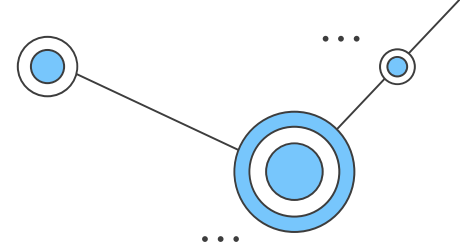




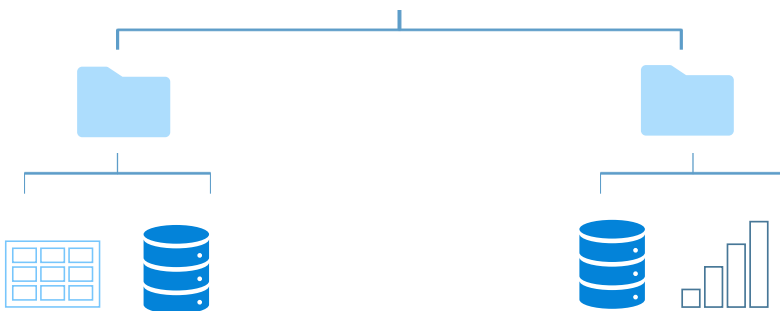
Subscription and management groups



Subscription and management groups



Subscription 1



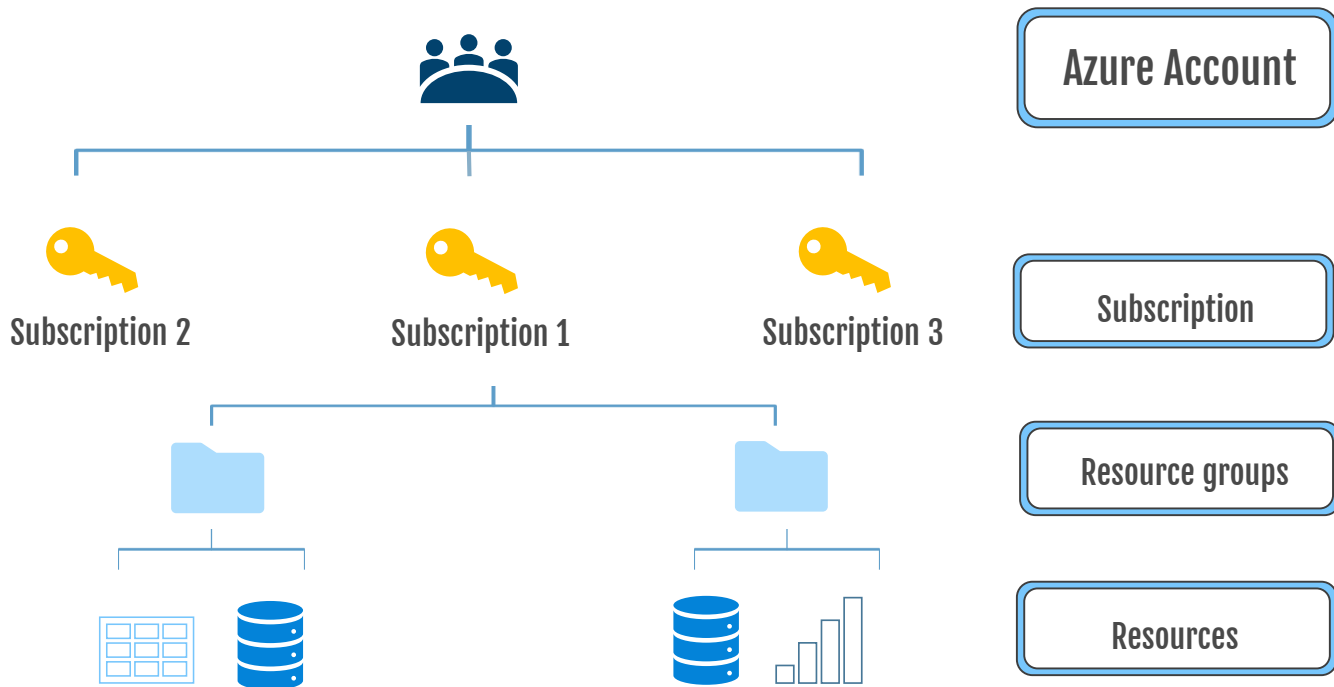
Azure Account

Subscription

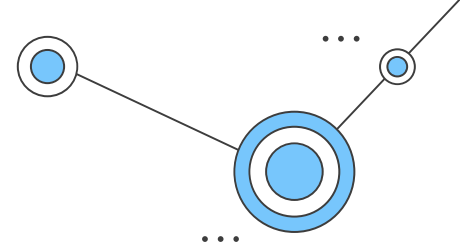
Resource groups

Resources

Subscription and management groups



Subscription and management groups



Azure Account

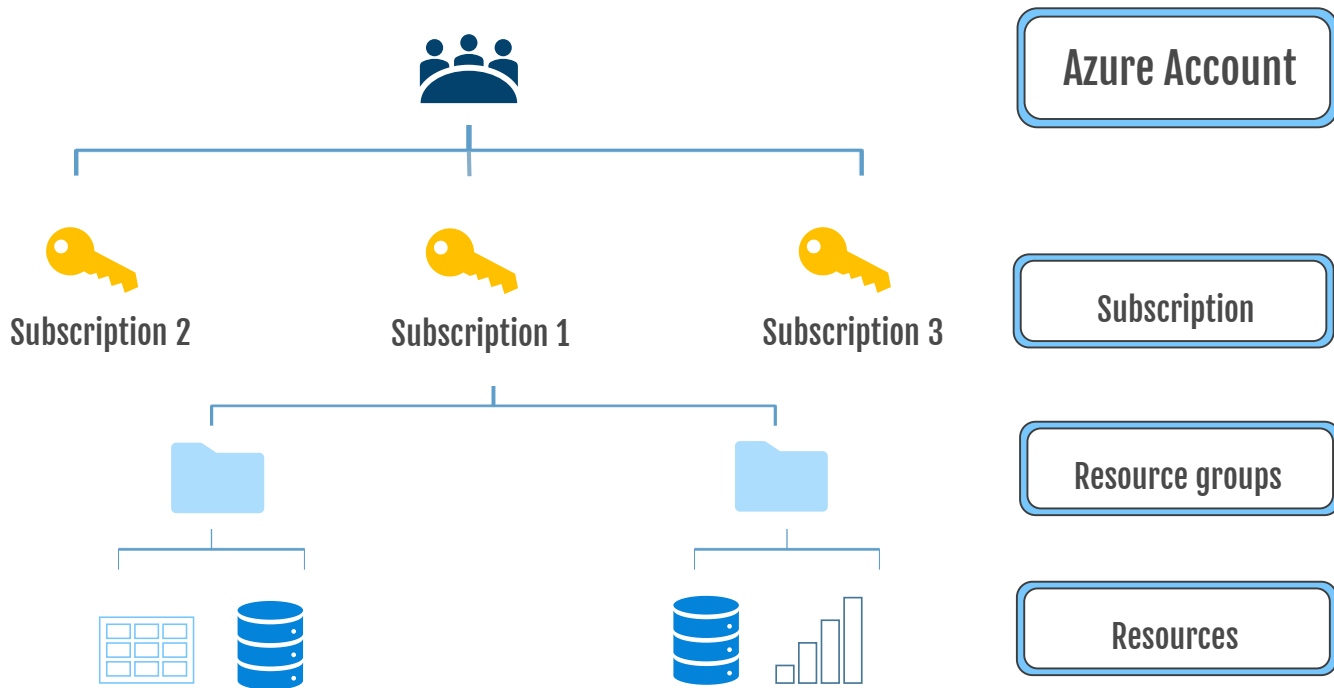
Subscription

Resource groups

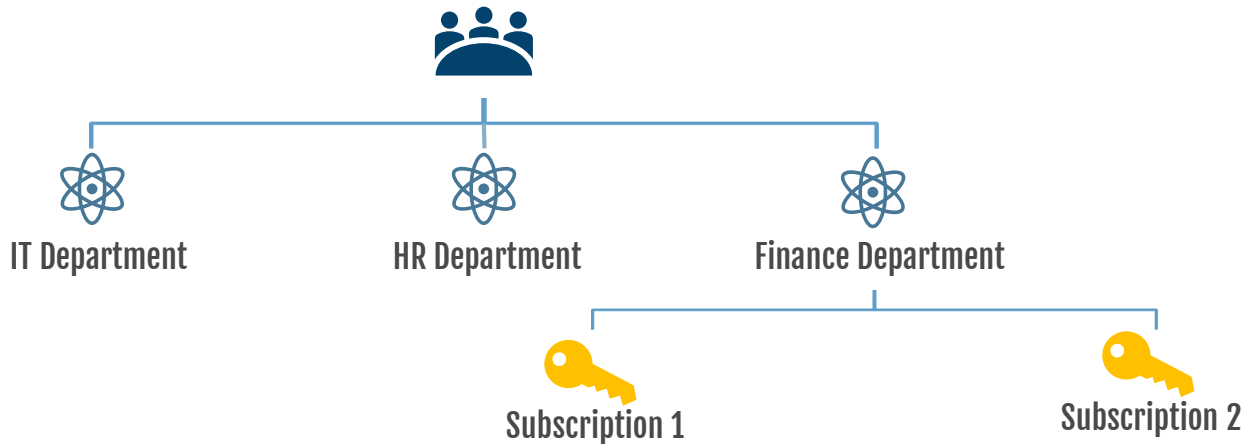
Resources



Subscription and management groups



Subscription and management groups



Azure Account

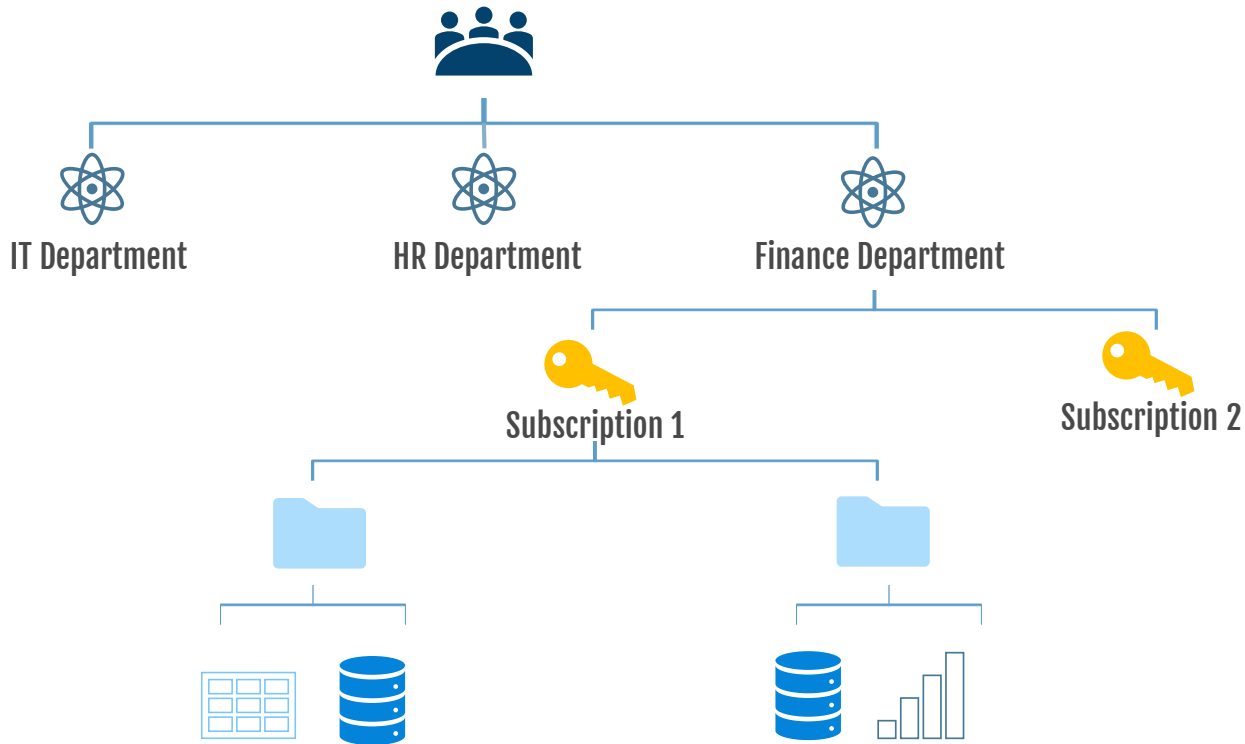
Management groups

Subscriptions

Resource groups

Resources

Subscription and management groups



Azure Account

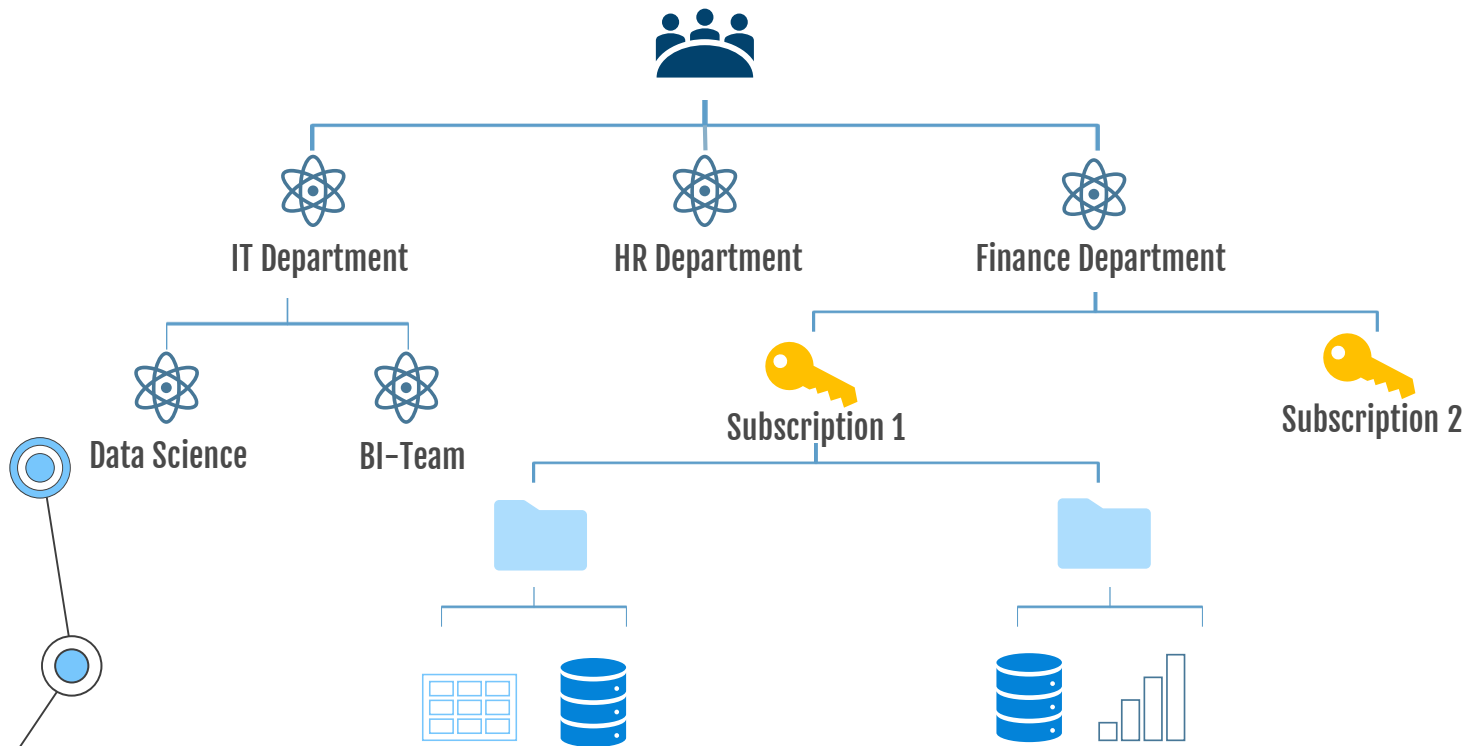
Management groups

Subscriptions

Resource groups

Resources

Subscription and management groups



Azure Account

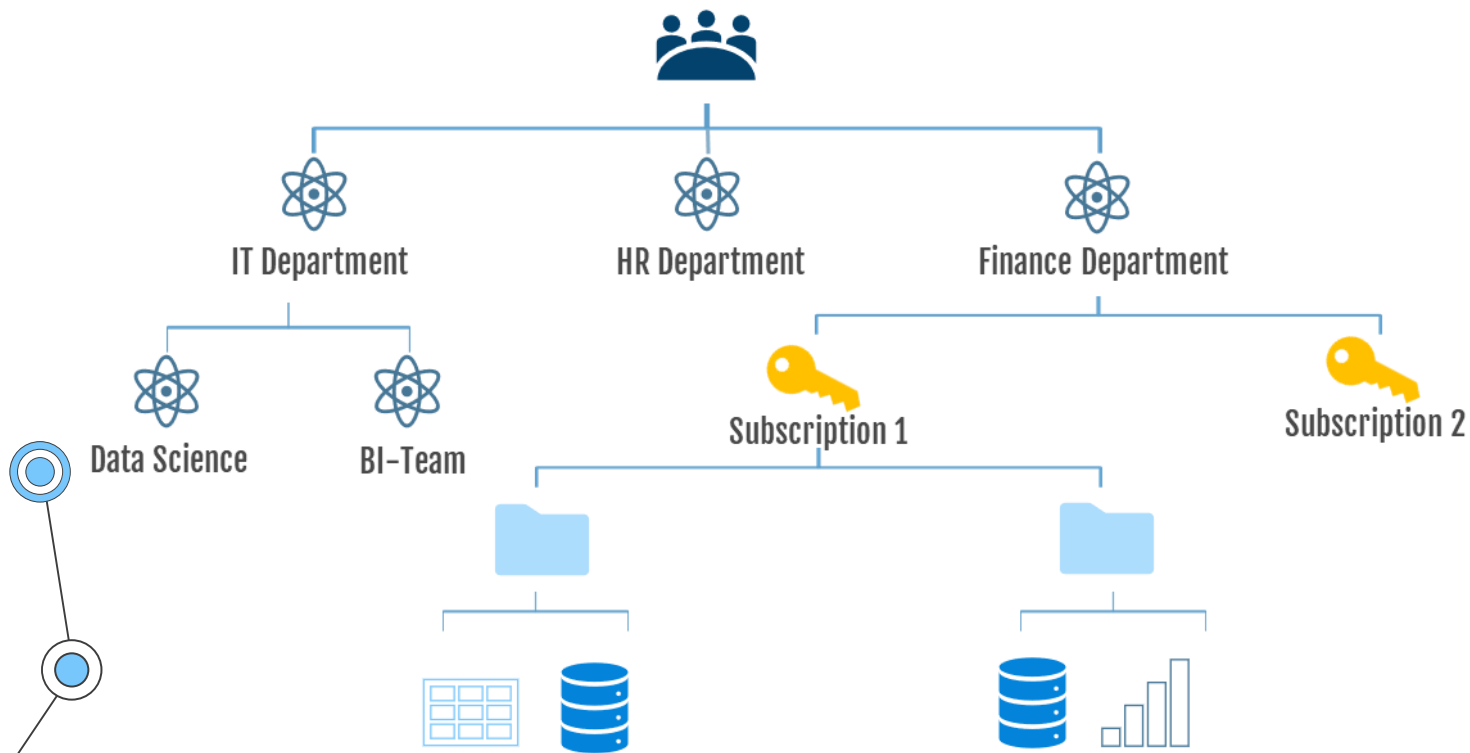
Management groups

Subscriptions

Resource groups

Resources

Subscription and management groups



Azure Account

Management groups

Subscriptions

Resource groups

Resources

Subscription and management groups

Resources

e.g. databases, virtual machines,
blob storage etc.
Can be moved to other subscriptions

Resource groups

Management of resources

Subscriptions

Account can have multiple
subscriptions
This is where billing takes place
Cannot be merged

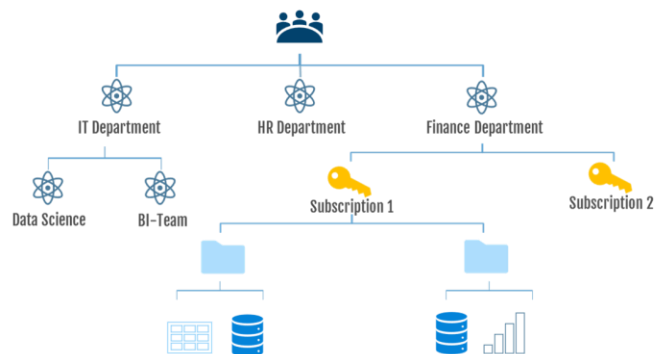
Management groups

Management of subscriptions &
policies
Can be nested

Environment: Test, Dev, Prod

Organizational structure

Billing purposes



Azure Account

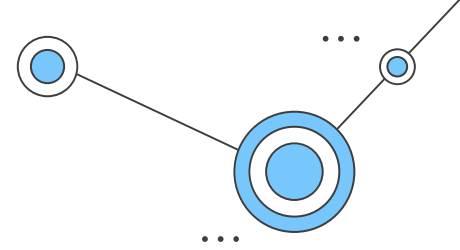
Management groups

Subscriptions

Resource groups

Resources

Summary



Geography

Area in the world, at least one region, define own market, data residency and compliance boundaries preserved

Region pairs

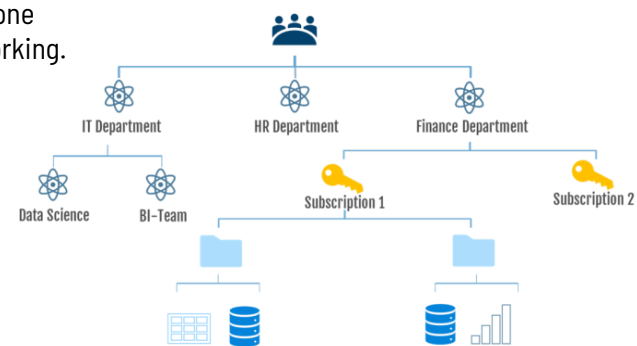
Two regions from the same geography

Region

Multiple data centers are connected within a radius via a dedicated regional network with low latency/latency.

Availability Zone

Physical locations within a region consisting of at least one data center with independent power, cooling, and networking.



Azure Account

Management groups

Subscriptions

Resource groups

Resources

Summary

Resources

e.g. databases, virtual machines, blob storage etc.

Resource groups

Management of resources

Subscriptions

This is where billing takes place

Management groups

Governance (e.g. via policies) across subscriptions

